

Education

Expected: 2030 PhD, Machine Learning

Georgia Institute of Technology, Atlanta, GA

Expected: 2027 MS, Electrical and Computer Engineering

Georgia Institute of Technology, Atlanta, GA

May 2026 BS, Software Engineering, *Summa cum laude*

Embry-Riddle Aeronautical University, Prescott, AZ

Recipient of *College of Engineering Outstanding Graduate Award*

GPA 4.0/4.0

May 2026 BS, Data Science, *Summa cum laude*

Embry-Riddle Aeronautical University, Prescott, AZ

Recipient of *Chancellor's Award* (highest distinction bestowed upon a graduating senior)

GPA 4.0/4.0

Research Interests

Machine learning, optimization, networked systems, online decision making, change-point detection.

Awards & Honors

- 2026** Chancellor's Award (Embry-Riddle Aeronautical University)
(Awarded by Embry-Riddle Aeronautical University to the most outstanding graduating senior)
- 2026** College of Engineering Outstanding Graduate Award (Embry-Riddle Aeronautical University)
- 2026** Ten Outstanding Young Persons Award (Junior Chamber International, Varese Chapter)
- 2026** IEEE-HKN Top Graduate Award
- 2026** Phi Kappa Phi Top Scholar Award
- 2026** 20 Twenties Award (Aviation Week Network by Accenture)
- 2023** Outstanding Student Award (West Virginia University Institute of Technology)

Research Projects

Graduate Projects

- 1.** Quickest Change Detection in Correlated Networks
Institution: Georgia Institute of Technology
PI: Matthew Hale
Role: Graduate Research Assistant
Project Dates: 05/2026 – Ongoing

Undergraduate Projects

- 1.** Modeling the Bonding Dynamics of Substrate Transferred Epitaxial Coating Films
Sponsor: National Science Foundation (NSF)
Institution: Embry-Riddle Aeronautical University
PI: Andri M. Gretarsson
Role: Undergraduate Research Assistant
Project Dates: 01/2026 – 08/2026
- 2.** Entropic Framework for Complex Network Structure Recovery
Sponsor: Undergraduate Research Institute (URI)
Institution: Embry-Riddle Aeronautical University
Advisor: Abd AlRahman Rasheed AlMomani
Role: Lead
Total Award: \$1,000
Project Dates: 09/2025 - 05/2026
- 3.** Passive Military Drone Detection Systems
Sponsor: Center for Aerospace Resilient Systems (CARS)
Institution: Embry-Riddle Aeronautical University
Advisor: Ahmed I. Sulyman
Role: Capstone Project Team Lead
Total Award: \$2,500

Project Dates: 08/2025 - 05/2026

- 4. North American Monsoon Prediction Using Causality Informed Machine Learning**
Sponsor: National Science Foundation (NSF)
Institution: Embry-Riddle Aeronautical University
Advisor: Curtis James
Role: Undergraduate Research Assistant
Project Dates: 01/2026 - 03/2026
- 5. A Geometric Aware Measure of Clustering Balance**
Institution: Embry-Riddle Aeronautical University
Advisors: C. Tyler Diggans, Abd AlRahman Rasheed AlMomani
Role: First Author
Project Dates: 10/2025 - 12/2025
- 6. Uncovering Direct Influence Networks of Takeoff Weight**
Sponsor: Undergraduate Research Institute (URI)
Institution: Embry-Riddle Aeronautical University
Role: Lead
Total Award: \$6,000
Project Dates: 09/2024 - 06/2025
- 7. SURF: Don't Bother the Driver: Sensor-Scheduling for Cognitive State Estimation During Automated Driving**
Sponsor: Engineering Undergraduate Research Office (EURO), National Science Foundation (NSF)
Institution: Purdue University
Role: Visiting Research Assistant
Total Award: \$10,000
Project Dates: 05/2025 - 08/2025
- 8. Human-AI Teaming: Measuring Human Trust in AI in Safety-Critical Systems**
Sponsor: Undergraduate Research Institute (URI)
Institution: Embry-Riddle Aeronautical University
Advisor: Timothy King
Role: Research Team Lead
Total Award: \$3,900
Project Dates: 09/2024 - 06/2025
- 9. AI-Based Phishing and Spam Detection**
Sponsor: Undergraduate Research Institute (URI)
Institution: Embry-Riddle Aeronautical University
Advisor: Timothy King
Role: Software Lead
Total Award: \$3,529
Project Dates: 04/2024 - 04/2025

Publications

Journal Papers

In Preparation

- J2** Bossi, E., Gretarsson, A.M. (2026). Modeling the Bonding Dynamics of Substrate Transferred Epitaxial Coating Films. In Preparation.

Submitted

- J1** Bossi, E., Diggans, C. Tyler, AlMomani, R. (2026). Boltzmann-Shannon Index: A Geometric-Aware Measure of Clustering Balance. Preprint at: <https://arxiv.org/abs/2512.02397>.

Conference Papers

Published and Accepted

- C4** Bossi, E., Jeevanandam S., Jain, N. (2026). Identifying Informative Physiological Indicators and Time Scales for Inferring Human Cognition During Automated Driving: An Information-Theoretic Approach. In ASPIRE 2026 - The HFES Annual Meeting and Exhibition.
- C3** Bossi, E., Jeevanandam S., Jain, N. (2026). A Sensor-Scheduling Approach to Predict Human Reliance on Automation During Automated Driving. In 2026 IFAC World Congress.
- C2** Bossi, E., AlMomani, R. (2026). Uncovering Direct Influence Networks of Takeoff Weight: Network Science and Information Theoretic Approach. In 2026 AIAA Science & Technology Forum.
- C1** Bossi, E., Ahmed, F. (2025). Enhancing Sentiment Analysis with Feature Extraction and Dimensionality Reduction in Traditional Machine Learning Models. In Intelligent Systems Conference (pp. 166-183). Cham: Springer Nature Switzerland.

Pre-Print

- P1** Bossi, E. (2026). ERpy: Python Implementation of Entropic Regression for Nonlinear System Identification. In Preparation.

Teaching

Embry-Riddle Aeronautical University

- Spring 2026** CI 460: Machine Learning & Big Data Analytics.
Teaching Assistant.
- Spring 2026** SE 300: Software Engineering Practices.
Teaching Assistant.
- Fall 2025** CI 460: Machine Learning & Big Data Analytics.
Teaching Assistant.
- Fall 2025** CS 420: Operating Systems.
Teaching Assistant.
- Fall 2025** PS 253: Physics III for Engineers Laboratory.
Laboratory Assistant.
- Spring 2025** CI 460: Machine Learning & Big Data Analytics.
Teaching Assistant.
- Spring 2025** SE 300: Software Engineering Practices.
Teaching Assistant.
- Fall 2024** CI 460: Machine Learning & Big Data Analytics.
Teaching Assistant.

Presentations and Invited Talks

- 2026** Presented the poster “An Entropic Framework For Sparse Network Reconstruction In Nonlinear Dynamical Systems” at the Embry-Riddle annual research symposium, Prescott, AZ, Mar., 2026. Winner of the Third Place “*Best Poster*” Award.
- 2026** Presented the paper “Uncovering Direct Influence Networks of Takeoff Weight: Network Science and Information Theoretic Approach” at the 2026 AIAA Science & Technology Forum, Orlando, FL, Jan., 2026.
- 2025** Presented the talk “The Future Is Now: How AI Is Reshaping Our World” at the Embry-Riddle Aeronautical University AI Summit, Prescott, AZ, Nov., 2025.
- 2025** Presented the paper “Enhancing Sentiment Analysis with Feature Extraction and Dimensionality Reduction in Traditional Machine Learning Models” at the XI Intelligent Systems Conference (IntelliSys), Amsterdam, The Netherlands, Aug., 2025.
- 2025** Presented the talk “Don’t Bother the Driver: Sensor-Scheduling for Cognitive State Estimation During Automated Driving” at the Purdue University Summer Research Symposium, West Lafayette, IN, Jul., 2025.
- 2025** Presented the talk “Measuring Human Trust in Artificial Intelligence in Safety-Critical Systems” at the Arizona Space Grant Symposium, Scottsdale, AZ, Apr., 2025.
- 2025** Presented the talk “AI-Driven Smart Agriculture for Climate Resilience” at the Global Stage of Invent for The Planet (IFTP), College Station, TX, Apr., 2025.
- 2025** Presented the talk “Phishing in the Digital Age: Surveying Public Awareness and Leveraging AI for Defense” at the 2025 National Conference on Undergraduate Research (NCUR), Pittsburgh, PA, Apr., 2025.
- 2025** Presented work on machine learning estimation of aircrafts takeoff weight at the Prescott Regional Science & Technology Fest, Prescott, AZ, Mar., 2025.

Service and Professional Activities

Professional Organization Memberships

- Member** The Order of The Engineer, Italian Scientists and Scholars in North America Foundation (ISSNAF), IEEE Eta Kappa Nu, Tau Beta Pi, Phi Kappa Phi.

Campus Involvement at Embry-Riddle Aeronautical University

- President & Founder** Data Science Club (2025 – 2026).

Varsity Athlete Men's Soccer (2022 – 2025).
Volunteer Men's Soccer (Spring 2026).
Assistant Coach